Dialectica over a Topos

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Abstract

We consider the dialectica categories Dial($\mathcal{E}$) and Gir($\mathcal{E}$) over a topos $\mathcal{E}$, with relations given by subobjects, that is, valued in the subobject classifier Ω. Over a non-Boolean topos these categories are more linear than over sets. Contraction by cases needs decidable relations, and we exhibit a sheaf topos with an object that has no map to its tensor square; decidability is sufficient for contraction but not necessary. Weakening needs global sections, and fails in Sh(S^1) for an object whose counterexamples are internally inhabited. In Gir($\mathcal{E}$) the negation only gives a map $A \to A^{\perp\perp}$. The modality ! with finite multisets of counterexamples exists over any topos with a natural numbers object and is still cofree, so the non-linearity it provides is untouched. The parametrised natural numbers objects keep their structural maps, so the topos does not by itself give a linear natural numbers object.

1 Introduction

The dialectica categories model linear logic, and the modality ! is meant to carry all the copying and discarding that intuitionistic logic does. Over sets this is not quite the whole story. Every object $A = (U, X, \alpha)$ of Dial(Sets) has a map $A \to A \otimes A$, by an argument by cases that Martin Hyland pointed out, and every object with a function $U \to X$ has a map to the unit. These maps are not natural and do not make A a comonoid, but their existence shows that a good part of the non-linearity of Dial(Sets) comes from the base: from the truth values being two, and from every non-empty set having an element. The question of this note is what happens to these maps when the base is a topos and the relations take values in Ω.

The answer is that both maps can fail, for different reasons. Contraction by cases needs a decision on whether u is related to x, and weakening needs a global section, which a sheaf can lack even when it is locally inhabited everywhere. The modality, on the other hand, needs neither: its contraction is union of multisets and its weakening is the empty multiset. We check that it exists and is cofree over any topos with a natural numbers object, by the argument of the thesis [1] read in the internal language. The Girard variant Gir($\mathcal{E}$) loses the involutive negation, as in the constructible falsity paper with José Siqueira [2], and we take this as a feature. Finally we check that the natural numbers objects of the joint note with Charles Morgan and Samuel da Silva [5] still copy, so that changing the base does not produce a linear natural numbers object.

2 Dialectica categories over a topos

This section fixes the definitions over a topos and records that the monoidal closed structure of the thesis is available. Throughout, $\mathcal{E}$ is a topos with a natural numbers object. We reason in the internal language of $\mathcal{E}$ [4, 3], which is intuitionistic. For subobjects φ, ψ of the same

object we write that $\varphi \Rightarrow \psi$ holds to mean $\varphi \leq \psi$ in the subobject lattice, and free variables are universally quantified. An argument by cases on a formula φ is valid when $\varphi \vee \neg\varphi$ holds, that is, when φ is decidable (complemented).

Definition 2.1. The category $\mathrm{Dial}(\mathcal{E})$ has as objects triples $A = (U, X, \alpha)$ with α a subobject of $U \times X$. A morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ is a pair of maps $f\colon U \to V$ and $F\colon U \times Y \to X$ of $\mathcal{E}$ such that

$$\alpha(u, F(u, y)) \Rightarrow \beta(fu, y)$$

holds. Composition is $(g, G) \circ (f, F) = (gf,\ (u, z) \mapsto F(u, G(fu, z)))$. The category $\mathrm{Gir}(\mathcal{E})$ has the same objects, and its morphisms are pairs $f\colon U \to V$, $F\colon Y \to X$ such that $\alpha(u, Fy) \Rightarrow \beta(fu, y)$ holds, composed componentwise.

We call U the witnesses and X the counterexamples, as over sets. The tensor of $\mathrm{Dial}(\mathcal{E})$ is $A \otimes B = (U \times V, X \times Y, \alpha \times \beta)$ with $(\alpha \times \beta)((u, v), (x, y)) = \alpha(u, x) \wedge \beta(v, y)$, and its unit is $I = (1, 1, \top)$. The tensor of $\mathrm{Gir}(\mathcal{E})$ is

$$A \otimes_{\mathrm{Gir}} B = (U \times V,\ X^V \times Y^U,\ \alpha(u, \varphi v) \wedge \beta(v, \psi u)),$$

with the same unit. The internal homs are

$$A \multimap B = (V^U \times X^{U \times Y},\ U \times Y,\ \alpha(u, F(u, y)) \Rightarrow \beta(fu, y))$$

in $\mathrm{Dial}(\mathcal{E})$, and the same with X^Y in place of $X^{U \times Y}$ and $\alpha(u, Fy)$ in place of $\alpha(u, F(u, y))$ in $\mathrm{Gir}(\mathcal{E})$. The linear falsum is $\bot = (1, 1, 0)$, with 0 the least subobject.

Proposition 2.2. *Both* $\mathrm{Dial}(\mathcal{E})$ *and* $\mathrm{Gir}(\mathcal{E})$ *are symmetric monoidal closed.*

Proof. For $\mathrm{Dial}(\mathcal{E})$, a morphism $A \otimes B \to C$, with $C = (W, Z, \gamma)$, is a map $h\colon U \times V \to W$ together with maps $H_1\colon U \times V \times Z \to X$ and $H_2\colon U \times V \times Z \to Y$ such that $\alpha(u, H_1(u, v, z)) \wedge \beta(v, H_2(u, v, z)) \Rightarrow \gamma(h(u, v), z)$. A morphism $A \to (B \multimap C)$ is a map $U \to W^V \times Y^{V \times Z}$ together with a map $U \times V \times Z \to X$ such that $\alpha(u, H_1(u, v, z)) \Rightarrow (\beta(v, H_2(u, v, z)) \Rightarrow \gamma(h(u, v), z))$. Currying in $\mathcal{E}$ matches the maps, and $\wedge \dashv \Rightarrow$ in $\mathrm{Sub}(U \times V \times Z)$ matches the conditions; naturality is that of currying. The case of $\mathrm{Gir}(\mathcal{E})$ is the same computation with the backward maps not depending on the witnesses. The coherence conditions hold because they hold in $\mathcal{E}$ and in each subobject lattice. □

No argument by cases is used here, so this part of the thesis holds over any topos. The category $\mathrm{Dial}(\mathcal{E})$ has products $A \times B = (U \times V, X + Y, \mathrm{ch})$, with ch choosing α or β according to the summand, and a terminal object $(1, 0, 0)$; the map $A \to (1, 0, 0)$ has as backward component the unique map $U \times 0 \cong 0 \to X$.

3 Contraction

Over sets every object has a map to its tensor square. This section shows that the usual proof needs decidable relations, that over a non-Boolean topos the map can fail to exist, and that decidability is nevertheless not necessary for it.

Proposition 3.1. *If α is decidable then $A = (U, X, \alpha)$ has a morphism $(\Delta, F)\colon A \to A \otimes A$ in* $\mathrm{Dial}(\mathcal{E})$*, where Δ is the diagonal and*

$$F(u, x, x') = \begin{cases} x' & \text{if } \alpha(u, x), \\ x & \text{if } \neg\alpha(u, x). \end{cases}$$

Proof. Since α is complemented, $U \times X \cong \alpha + \neg\alpha$, and as products distribute over coproducts in a topos, $U \times X \times X \cong (\alpha \times X) + (\neg\alpha \times X)$. The map F is the projection to the second copy of X on the first summand and the projection to the first copy on the second. We must show $\alpha(u, F(u, x, x')) \Rightarrow \alpha(u, x) \wedge \alpha(u, x')$, and since $\alpha(u, x) \vee \neg\alpha(u, x)$ holds we may argue by cases. If $\alpha(u, x)$ then $F(u, x, x') = x'$, and $\alpha(u, x')$ gives the conjunction. If $\neg\alpha(u, x)$ then $F(u, x, x') = x$, and the hypothesis $\alpha(u, x)$ is contradictory. □

Over sets every relation is decidable, which is why every object of Dial(Sets) has such a map. Before giving an object without one, note that some objects contract for reasons that have nothing to do with decidability.

Example 3.2. Any object $(U, 1, \alpha)$, with α an arbitrary subobject of U, has the morphism $(\Delta, !)\colon (U, 1, \alpha) \to (U, 1, \alpha) \otimes (U, 1, \alpha)$, where $!$ is the unique map to 1. The condition reads $\alpha(u) \Rightarrow \alpha(u) \wedge \alpha(u)$. Similarly any object $(U, 0, 0)$ has a map to its tensor square, since the backward component is the unique map out of $U \times 0 \times 0 \cong 0$.

The interesting case is therefore one where the counterexamples can be compared but not decided. We say $\mathcal{E}$ is connected when 1 has no complemented subobjects other than 0 and 1, so that the global elements of $2 = 1 + 1$ are the two coprojections, written 0 and 1. Given subobjects p, q of 1, let $A_{p,q} = (1, 2, p + q)$, the relation being p on the first summand of 2 and q on the second.

Proposition 3.3. *Let $\mathcal{E}$ be connected. There is a morphism $A_{p,q} \to A_{p,q} \otimes A_{p,q}$ in* $\mathrm{Dial}(\mathcal{E})$ *if and only if $p \leq q$ or $q \leq p$.*

Proof. Write $\alpha_0 = p$ and $\alpha_1 = q$. The forward component $1 \to 1 \times 1$ is unique. The backward component is a map $F\colon 1 \times 2 \times 2 \to 2$, and since $2 \times 2 \cong 1 + 1 + 1 + 1$ it is given by four global elements $F(i, j)$ of 2, each of which is 0 or 1 because $\mathcal{E}$ is connected. On the summand (i, j) the condition $\alpha(F) \Rightarrow \alpha(i) \wedge \alpha(j)$ becomes $\alpha_{F(i,j)} \leq \alpha_i \wedge \alpha_j$ in $\mathrm{Sub}(1)$. For $i = j$ the choice $F(i, i) = i$ satisfies it. For $(i, j) = (0, 1)$ the choice $F(0, 1) = 0$ requires $p \leq p \wedge q$, that is $p \leq q$, and the choice $F(0, 1) = 1$ requires $q \leq p$. The summand $(1, 0)$ gives the same two alternatives. So a suitable F exists exactly when $p \leq q$ or $q \leq p$. □

Example 3.4. Let $V = \{a, b, c\}$ with open sets $\emptyset, \{a\}, \{b\}, \{a, b\}, V$. Every open set containing c is V, so V is connected and so is $\mathrm{Sh}(V)$, whose truth values are the open sets of V. Take $p = \{a\}$ and $q = \{b\}$. They are incomparable, so by Proposition 3.3 the object $A_{\{a\},\{b\}}$ has no morphism to its tensor square in $\mathrm{Dial}(\mathrm{Sh}(V))$. On the other hand $A_{\{a\},\{a\}}$ has one, with F the first projection, although its relation is not decidable: the pseudo-complement of $\{a\}$ is $\{b\}$, and $\{a\} \cup \{b\} = \{a, b\} \neq V$. The same two examples can be given in $\mathrm{Sh}(\mathbb{R})$ with $p = (-\infty, 1)$ and $q = (0, \infty)$.

So the objects of $\mathrm{Dial}(\mathcal{E})$ admitting contraction are not exactly those with decidable relations. Decidability is sufficient, by Proposition 3.1, and it is not necessary, by Examples 3.2 and 3.4. For two-element sets of counterexamples over a connected topos the right condition is comparability of the two truth values, which over sets always holds.

4 Weakening

Over sets an object fails to have a map to the unit only when it has witnesses and no counterexamples. In this section we see that over a sheaf topos weakening fails much more often, including for objects whose counterexamples are internally inhabited.

Proposition 4.1. *In* $\mathrm{Dial}(\mathcal{E})$ *the morphisms* $A \to I$ *are in bijection with the maps* $U \to X$ *of* $\mathcal{E}$*. In* $\mathrm{Gir}(\mathcal{E})$ *they are in bijection with the global elements* $1 \to X$*.*

Proof. The forward component $U \to 1$ is unique. In $\mathrm{Dial}(\mathcal{E})$ the backward component is a map $U \times 1 \cong U \to X$, and in $\mathrm{Gir}(\mathcal{E})$ it is a map $1 \to X$. In both cases the condition reads $\alpha(u, F(\dots)) \Rightarrow \top$, which always holds. □

The relation plays no role, so whether A discards depends only on the objects U and X of $\mathcal{E}$. Over sets a map $U \to X$ exists unless X is empty and U is not. In a sheaf topos there may be no map $U \to X$ even when $\mathcal{E}$ believes that X is inhabited.

Example 4.2. Let $p\colon S^1 \to S^1$ be the double cover $z \mapsto z^2$, and let X be the sheaf on S^1 of continuous local square roots, $X(W) = \{s\colon W \to S^1 \text{ continuous} \mid s(z)^2 = z\}$. Since p is a covering map every point has a neighbourhood on which X has a section, so $X \to 1$ is an epimorphism and $\exists x \in X.\, \top$ holds in $\mathrm{Sh}(S^1)$. A global section would be a continuous square root on all of S^1, which does not exist. Hence the object $A = (1, X, \top)$ has no morphism $A \to I$ in $\mathrm{Dial}(\mathrm{Sh}(S^1))$, nor in $\mathrm{Gir}(\mathrm{Sh}(S^1))$, although internally its object of counterexamples is inhabited.

The failure is external: $A \to I$ asks for a map in $\mathcal{E}$, while the internal statement $\forall u\, \exists x$ only gives local choices. Any subterminal $W \neq 1$ used as counterexamples gives a simpler instance, but there X is not internally inhabited.

5 Negation in Gir

Over sets the negation of Gir is a contravariant involution. This section computes it over a topos, where it becomes the Heyting negation of Ω and only the map $A \to A^{\perp\perp}$ remains.

Proposition 5.1. *In* $\mathrm{Gir}(\mathcal{E})$*, with* $A^\perp = A \multimap \perp$*, we have* $A^\perp \cong (X, U, \neg\alpha^\circ)$*, where* $\alpha^\circ(x,u) = \alpha(u,x)$*, and hence* $A^{\perp\perp} \cong (U, X, \neg\neg\alpha)$*.*

Proof. By definition $A \multimap \perp = (1^U \times X^1,\ U \times 1,\ \alpha(u,x) \Rightarrow 0)$, and $1^U \cong 1$, $X^1 \cong X$, $U \times 1 \cong U$. Applying this twice gives the second statement. □

Proposition 5.2. *The pair* $(\mathrm{id}_U, \mathrm{id}_X)$ *is a morphism* $A \to A^{\perp\perp}$ *in* $\mathrm{Gir}(\mathcal{E})$*, and it is an isomorphism if and only if* $\alpha = \neg\neg\alpha$*.*

Proof. The condition is $\alpha(u,x) \Rightarrow \neg\neg\alpha(u,x)$, which holds intuitionistically. An inverse would have to be $(\mathrm{id}_U, \mathrm{id}_X)$ in the other direction, which is a morphism exactly when $\neg\neg\alpha \Rightarrow \alpha$ holds. □

Example 5.3. In $\mathrm{Sh}(\mathbb{R})$ take $A = (1, 1, \mathbb{R} \setminus \{0\})$. The pseudo-complement of $\mathbb{R} \setminus \{0\}$ is the interior of $\{0\}$, which is empty, so $A^{\perp\perp} = (1, 1, \top) = I$. The only candidate for a morphism $I \to A$ is $(\mathrm{id}, \mathrm{id})$, and it would require $\top \leq \mathbb{R} \setminus \{0\}$, so A and $A^{\perp\perp}$ are not isomorphic.

Thus $\mathrm{Gir}(\mathcal{E})$ is a model of intuitionistic linear logic in which every object maps to its double negation, and the classical duality of $\mathrm{Gir}(\mathrm{Sets})$ is recovered exactly on the $\neg\neg$-stable objects. This is the same one-way duality that José Siqueira and I used for constructible falsity over a Heyting algebra [2], here with Ω in the role of the Heyting algebra and $\mathcal{E}$ in the role of sets. We regard it as the expected behaviour of a linear negation over intuitionistic truth values, not as a defect.

6 The modality

Contraction and weakening failing for objects of $\mathrm{Dial}(\mathcal{E})$ would be a problem if they also failed for $!A$. This section shows that they do not: the modality of the thesis needs only free commutative monoids, which exist in a topos with a natural numbers object, and it is still the cofree commutative comonoid.

In $\mathcal{E}$ the list object X^{L} exists because $\mathcal{E}$ has a natural numbers object, and the free commutative monoid X^* is its quotient by the equivalence relation generated by permuting adjacent entries [3]. We write $\emptyset$ for the empty multiset, $[\![x]\!]$ for a singleton and $m + m'$ for union. Given $\alpha \leq U \times X$, define $\alpha_{\mathrm{L}} \leq U \times X^{\mathrm{L}}$ by list recursion, with $\alpha_{\mathrm{L}}(u, [\,]) = \top$ and $\alpha_{\mathrm{L}}(u, x :: l) = \alpha(u, x) \wedge \alpha_{\mathrm{L}}(u, l)$. Since $\wedge$ is commutative, α_{L} is stable under the equivalence relation, and since $U \times q$ is a regular epimorphism for the quotient map $q\colon X^{\mathrm{L}} \to X^*$, it is the pullback of a unique $\alpha^* \leq U \times X^*$. Internally $\alpha^*(u, m)$ says that $\alpha(u, x)$ holds for every x in m, and we have $\alpha^*(u, \emptyset) = \top$, $\alpha^*(u, [\![x]\!]) = \alpha(u, x)$ and $\alpha^*(u, m + m') = \alpha^*(u, m) \wedge \alpha^*(u, m')$.

Definition 6.1. For $A = (U, X, \alpha)$ in $\mathrm{Dial}(\mathcal{E})$ let $!A = (U, X^*, \alpha^*)$. Dereliction is $\varepsilon_A = (\mathrm{id}_U, (u, x) \mapsto [\![x]\!])\colon !A \to A$.

Proposition 6.2. *Every $!A$ is a commutative comonoid in $\mathrm{Dial}(\mathcal{E})$, with counit $(!, u \mapsto \emptyset)\colon !A \to I$ and comultiplication $(\Delta, (u, m, m') \mapsto m + m')\colon !A \to !A \otimes !A$.*

Proof. The condition for the counit is trivial. The condition for the comultiplication is $\alpha^*(u, m + m') \Rightarrow \alpha^*(u, m) \wedge \alpha^*(u, m')$, which holds with equality. The comonoid equations reduce, on the backward components, to $\emptyset$ being a unit for $+$ and $+$ being associative and commutative. Dereliction is a morphism because $\alpha^*(u, [\![x]\!]) = \alpha(u, x)$. □

No case split is used: union of multisets does the work that the argument by cases did in Proposition 3.1, and the empty multiset supplies the map $U \to X^*$ required by Proposition 4.1. So $!A$ copies and discards even when A is the object of Example 3.4 or of Example 4.2.

Theorem 6.3. *The object $!A$ with dereliction ε_A is the cofree commutative comonoid on A in $\mathrm{Dial}(\mathcal{E})$.*

Proof. Let $C = (W, Z, \gamma)$ be a commutative comonoid with counit $(!, e)$, $e\colon W \to Z$, and comultiplication (d, D), $d\colon W \to W \times W$, $D\colon W \times Z \times Z \to Z$. Composing with the projections, the forward parts of the counit equations give $\pi_1 d = \pi_2 d = \mathrm{id}$, so $d = \Delta$, and their backward parts give $D(w, e(w), z) = z = D(w, z, e(w))$. Coassociativity and cocommutativity give $D(w, D(w, z_1, z_2), z_3) = D(w, z_1, D(w, z_2, z_3))$ and $D(w, z_1, z_2) = D(w, z_2, z_1)$. Thus, internally, for each w the object Z is a commutative monoid with unit $e(w)$ and multiplication $D(w, -, -)$; externally, W^*Z is a commutative monoid in $\mathcal{E}/W$. The condition on the comultiplication is

$$\gamma(w, D(w, z_1, z_2)) \Rightarrow \gamma(w, z_1) \wedge \gamma(w, z_2).$$

Let $(g, G)\colon C \to A$ be a morphism. A comonoid morphism $(h, H)\colon C \to !A$ with $\varepsilon_A \circ (h, H) = (g, G)$ must have $h = g$, since ε_A is the identity on witnesses, and $H(w, [\![x]\!]) = G(w, x)$. Preserving the counit and comultiplication says, on backward components, $H(w, \emptyset) = e(w)$ and $H(w, m + m') = D(w, H(w, m), H(w, m'))$. So $H(w, -)$ must be the monoid homomorphism from X^* to $(Z, D(w, -, -), e(w))$ extending $G(w, -)$, and by the freeness of X^*, which is preserved by the logical functor $W^*\colon \mathcal{E} \to \mathcal{E}/W$, there is exactly one such map.

It remains to show that (g, H) is a morphism of $\mathrm{Dial}(\mathcal{E})$, that is, $\gamma(w, H(w, m)) \Rightarrow \alpha^*(gw, m)$. We prove this by induction on m, which is valid for arbitrary subobjects since it is induction on lists followed by descent to the quotient. For $m = \emptyset$ the right-hand side is $\top$. For $m = [\![x]\!]$ it is the condition $\gamma(w, G(w, x)) \Rightarrow \alpha(gw, x)$ on (g, G). For $m + m'$ we have

$$\gamma(w, H(w, m + m')) = \gamma(w, D(w, H(w, m), H(w, m'))) \Rightarrow \gamma(w, H(w, m)) \wedge \gamma(w, H(w, m')),$$

and the induction hypotheses give $\alpha^*(gw, m) \wedge \alpha^*(gw, m') = \alpha^*(gw, m + m')$. □

Promotion, the comultiplication $!A \to !!A$, is obtained from the theorem in the usual way, and is the map flattening a multiset of multisets. Girard's isomorphisms also hold: the product $A \times B$ has counterexamples $X + Y$, and $(X + Y)^* \cong X^* \times Y^*$ with $(\alpha + \beta)^*$ corresponding to $\alpha^* \times \beta^*$, which gives $!(A \times B) \cong !A \otimes !B$; and $0^* \cong 1$ with the relation $\top$, which gives $!(1, 0, 0) \cong I$. So the modality of the thesis lives entirely on the intuitionistic part of $\mathcal{E}$, and the change of base affects only the linear objects.

7 Natural numbers

The joint note with Charles Morgan and Samuel da Silva [5] finds a parametrised natural numbers object for each structure of a dialectica category: $(\mathbb{N}, 0, 0)$ for the products and $(\mathbb{N}, 1, \top)$ for the tensor. Its base is a cartesian closed category with finite limits, disjoint coproducts and a natural numbers object, whose subobject lattices are Heyting algebras with universal quantification along projections, stably under pullback. A topos with a natural numbers object is such a base, so both objects are parametrised natural numbers objects in $\mathrm{Dial}(\mathcal{E})$, with the verification of the dialectica condition done by induction in $\mathrm{Sub}(-)$ exactly as there. In this section we check whether the change of base makes either of them linear.

Proposition 7.1. *In* $\mathrm{Dial}(\mathcal{E})$ *the object* $(\mathbb{N}, 0, 0)$ *is a comonoid for the products, and* $(\mathbb{N}, 1, \top)$ *is a cocommutative comonoid for the tensor and a retract of* $!(\mathbb{N}, 1, \top)$.

Proof. Every object is a comonoid for the products, by the diagonal and the map to the terminal object $(1, 0, 0)$. For $(\mathbb{N}, 1, \top)$ the comonoid structure is given by the result of the joint note that any parametrised natural numbers object in the sense of Paré and Román is a cocommutative comonoid in any symmetric monoidal category, proved in the section on comonoids of [5]. It can also be read off directly: the counterexamples are 1, so the backward components of the maps to I and to the tensor square are unique, and the conditions are trivial, as in Example 3.2. For the retraction, the free commutative monoid on 1 in $\mathcal{E}$ is $\mathbb{N}$, so $!(\mathbb{N}, 1, \top) = (\mathbb{N}, \mathbb{N}, \top)$; the identity on witnesses with the unique backward map $\mathbb{N} \times \mathbb{N} \to 1$ is a morphism $(\mathbb{N}, 1, \top) \to !(\mathbb{N}, 1, \top)$, dereliction is a morphism back, and the composite is the identity, all conditions holding because the relations are full. □

Neither argument looks at the truth values: the relations are 0 and $\top$, and the counterexamples are 0 and 1, so the backward components are forced over any base. The object $(\mathbb{N}, 0, 0)$

has no morphism to I, since that would need a map $\mathbb{N} \to 0$, but this concerns the tensor, for which it was never a natural numbers object, and it holds over sets as well. So passing from sets to a non-Boolean topos makes some objects of $\mathrm{Dial}(\mathcal{E})$ linear but leaves the natural numbers as they were, and a linear natural numbers object needs a different universal property rather than a different base.

8 Conclusions

Over a topos the dialectica categories $\mathrm{Dial}(\mathcal{E})$ and $\mathrm{Gir}(\mathcal{E})$ keep their symmetric monoidal closed structure, and the modality ! of the thesis keeps its universal property, with proofs that use no argument by cases. What changes is the linear part. Contraction by cases is available for objects with decidable relations, but not for all objects: in $\mathrm{Dial}(\mathrm{Sh}(V))$ there is an object with no map to its tensor square, while some objects with undecidable relations still contract. Weakening needs a map from witnesses to counterexamples in $\mathcal{E}$, and fails in $\mathrm{Sh}(S^1)$ for an object whose counterexamples are internally inhabited. The negation of $\mathrm{Gir}(\mathcal{E})$ is the Heyting negation, with a map $A \to A^{\perp\perp}$ that is invertible exactly on the $\neg\neg$-stable objects. So over a non-Boolean topos the dialectica categories are more linear than over sets, and the non-linearity that remains is the one carried by !. The parametrised natural numbers objects of the joint note exist over $\mathcal{E}$ and are not affected: their relations and counterexamples are trivial, so they copy over any base.

References

[1] V. de Paiva. *The Dialectica Categories*. PhD thesis, University of Cambridge. Computer Laboratory Technical Report 213, 1991.

[2] V. de Paiva and J. Siqueira. Dialectica for constructible falsity. arXiv preprint.

[3] P. T. Johnstone. *Sketches of an Elephant: A Topos Theory Compendium*. Oxford University Press, 2002.

[4] S. Mac Lane and I. Moerdijk. *Sheaves in Geometry and Logic*. Springer, 1992.

[5] V. de Paiva, C. Morgan and S. G. da Silva. Parametrised Recursion in Dialectica Categories. Draft, 2026.